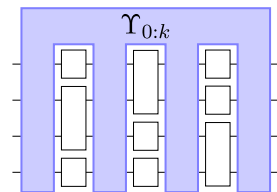
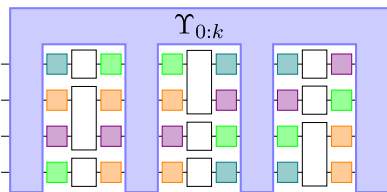


(b) Bridge: Process tensors to spatiotemporal Pauli processes

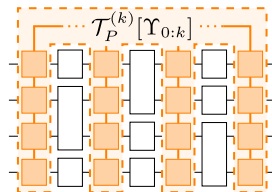
Process tensor noise model



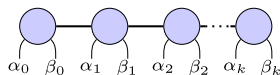
Multi-time Pauli twirl



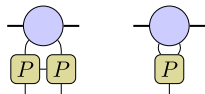
Spatiotemporal Pauli process



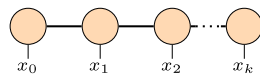
Process tensor MPO



Local Pauli twirl contraction



SPP MPS

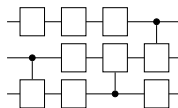


(a) Open quantum systems

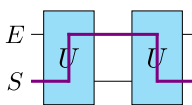
System-environment dynamics

$$\mathcal{H} = \mathcal{H}_S + \mathcal{H}_E + \mathcal{H}_{SE}$$

QEC circuit

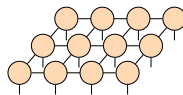


Memory effects



(c) Analysis and simulation

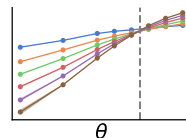
SPP tensor networks



Transfer operator

$$T = \sum_x A_x$$

QEC simulation



Hidden Markov models

